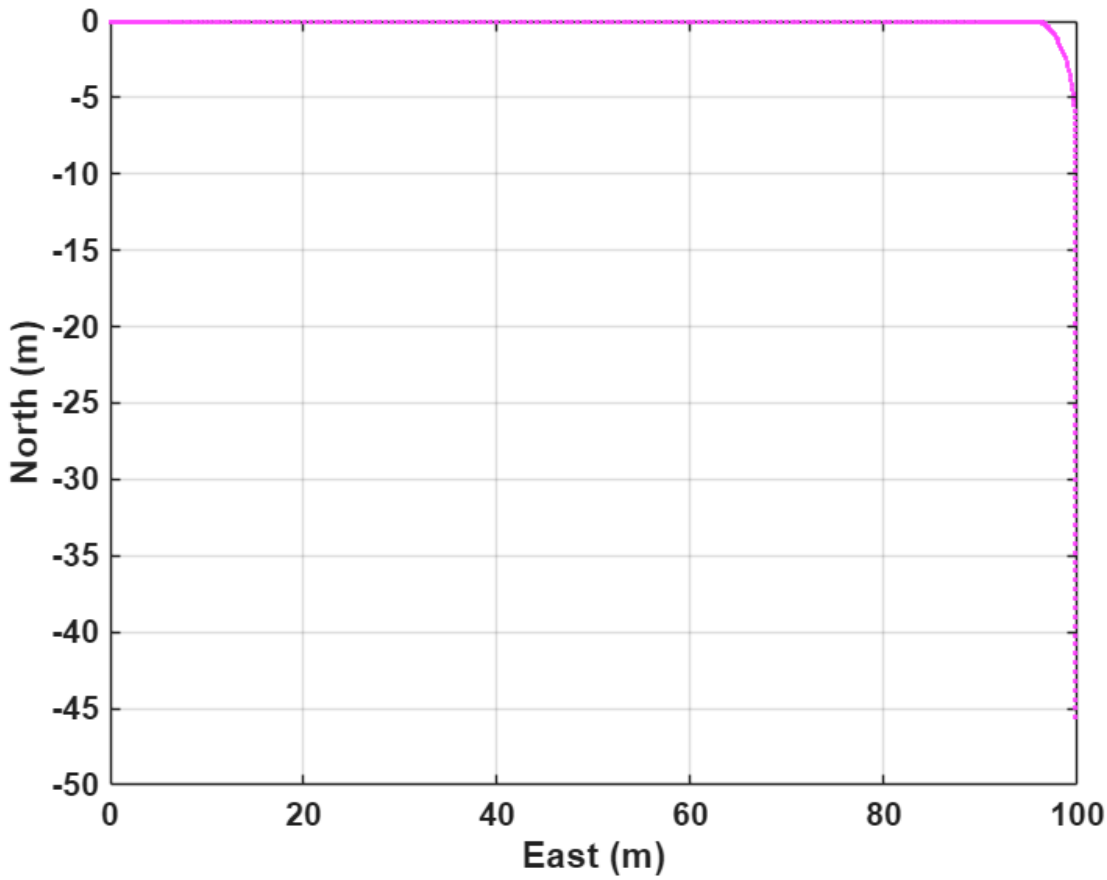
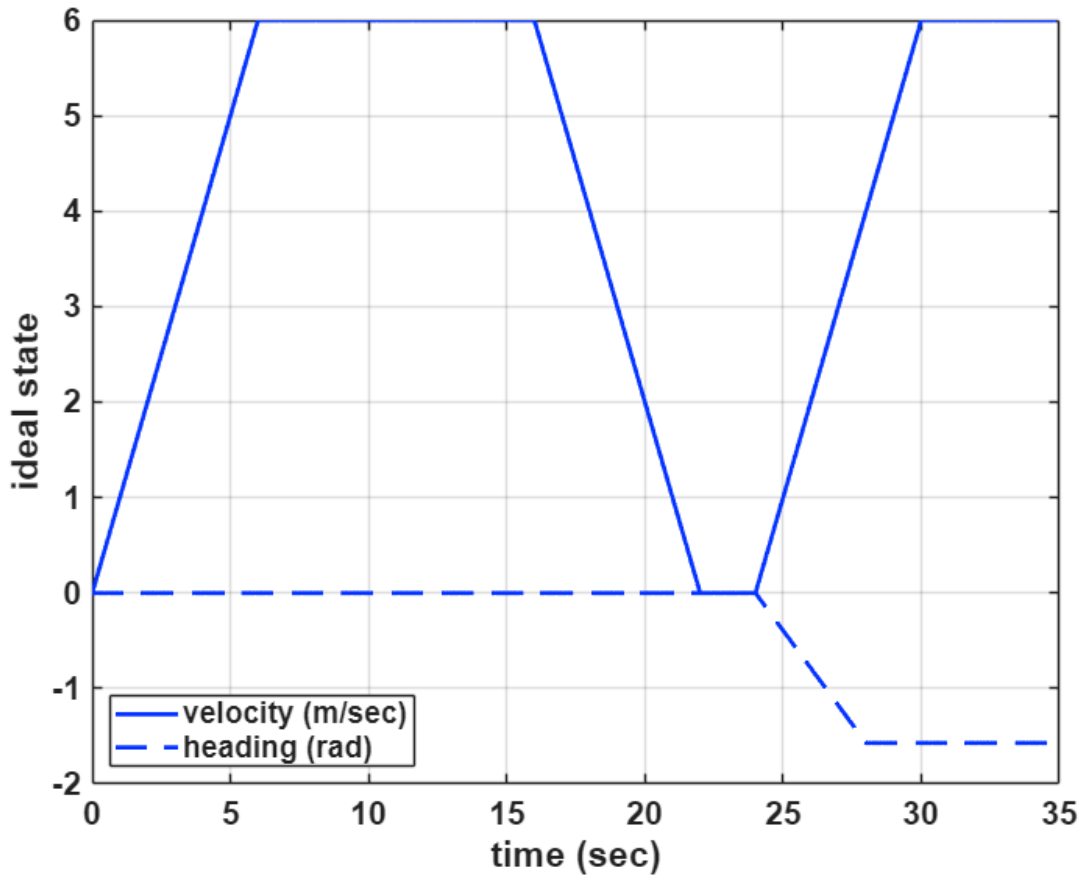
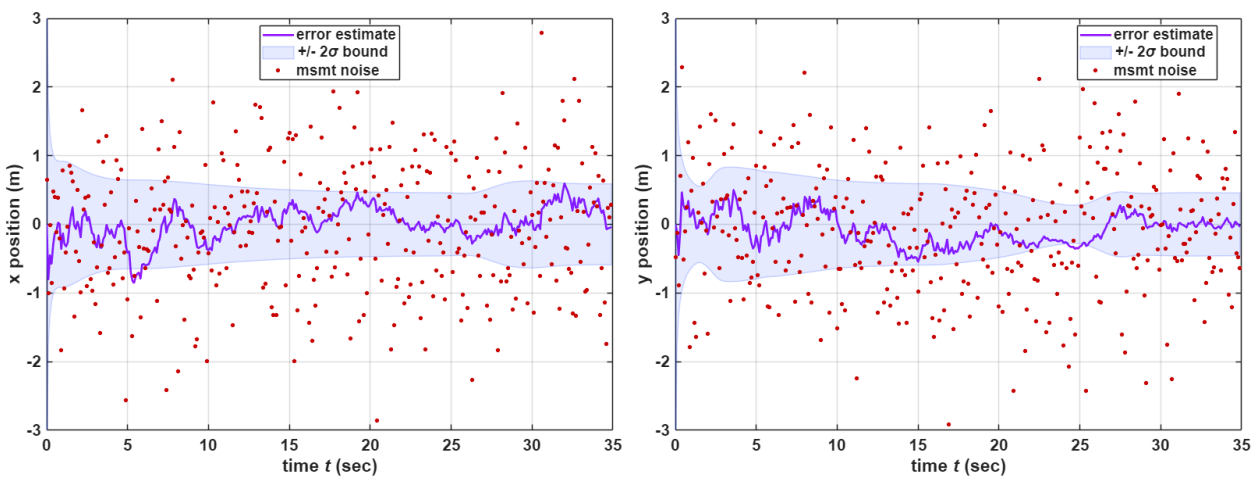
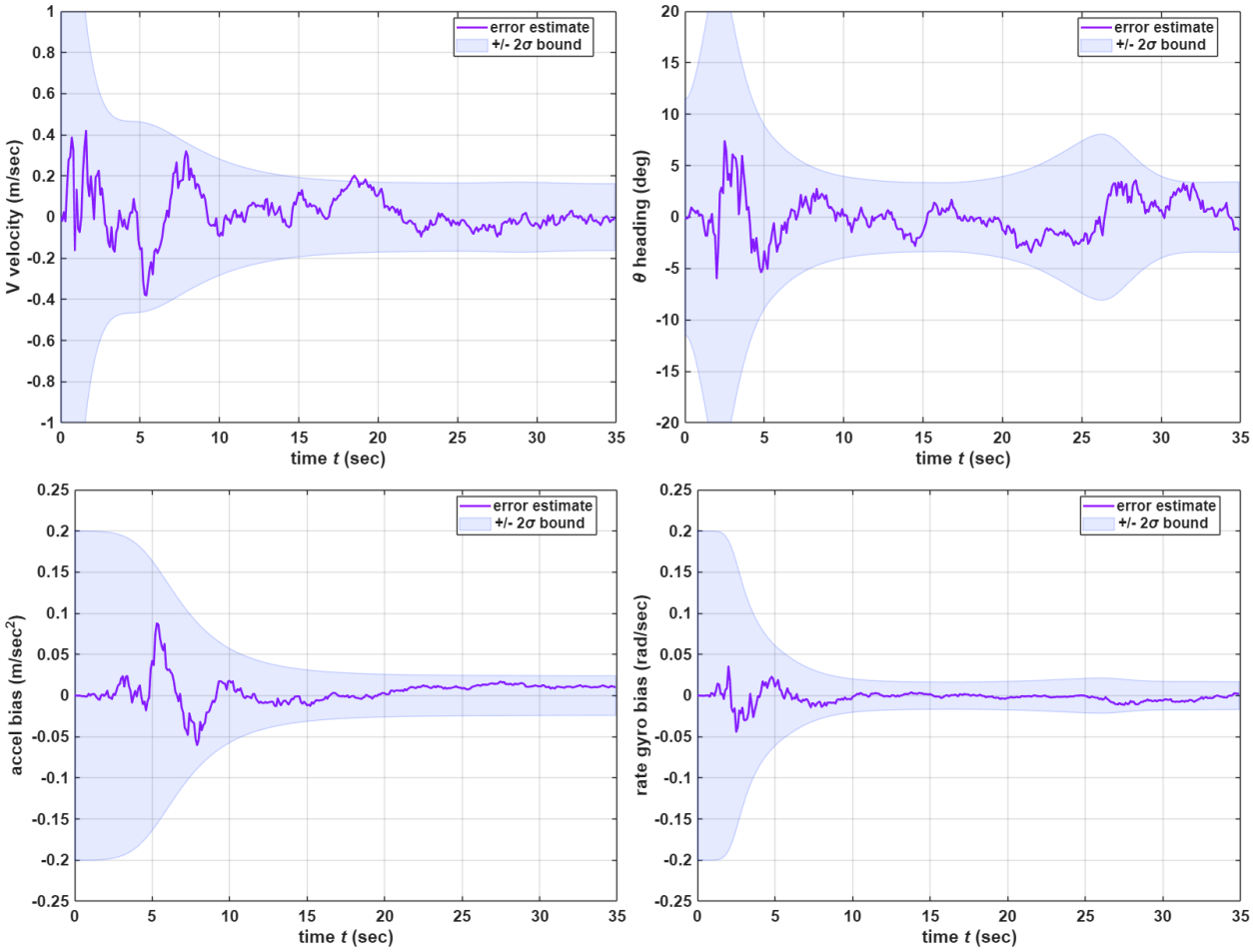
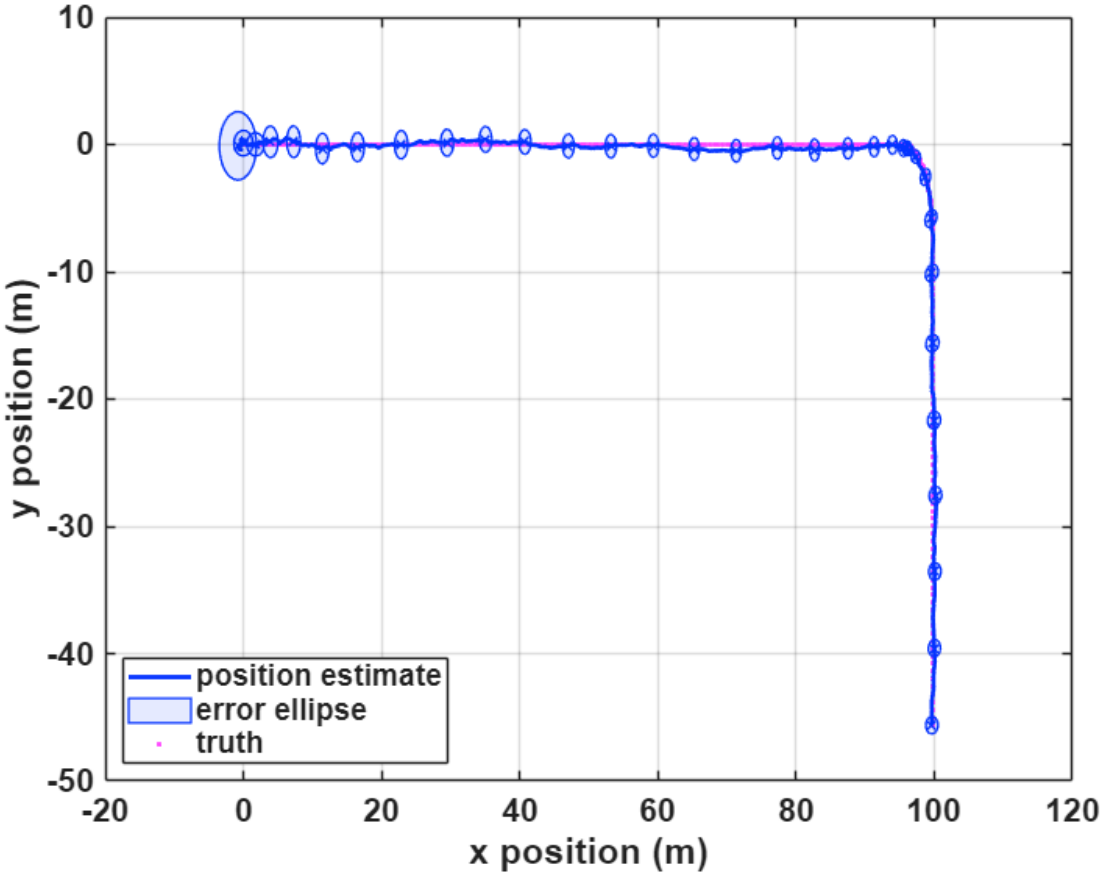
```
hw4_problem1
hw4_problem1_swervy

hw4_problem1_no_bias
hw4_problem1_no_bias_swervy
```

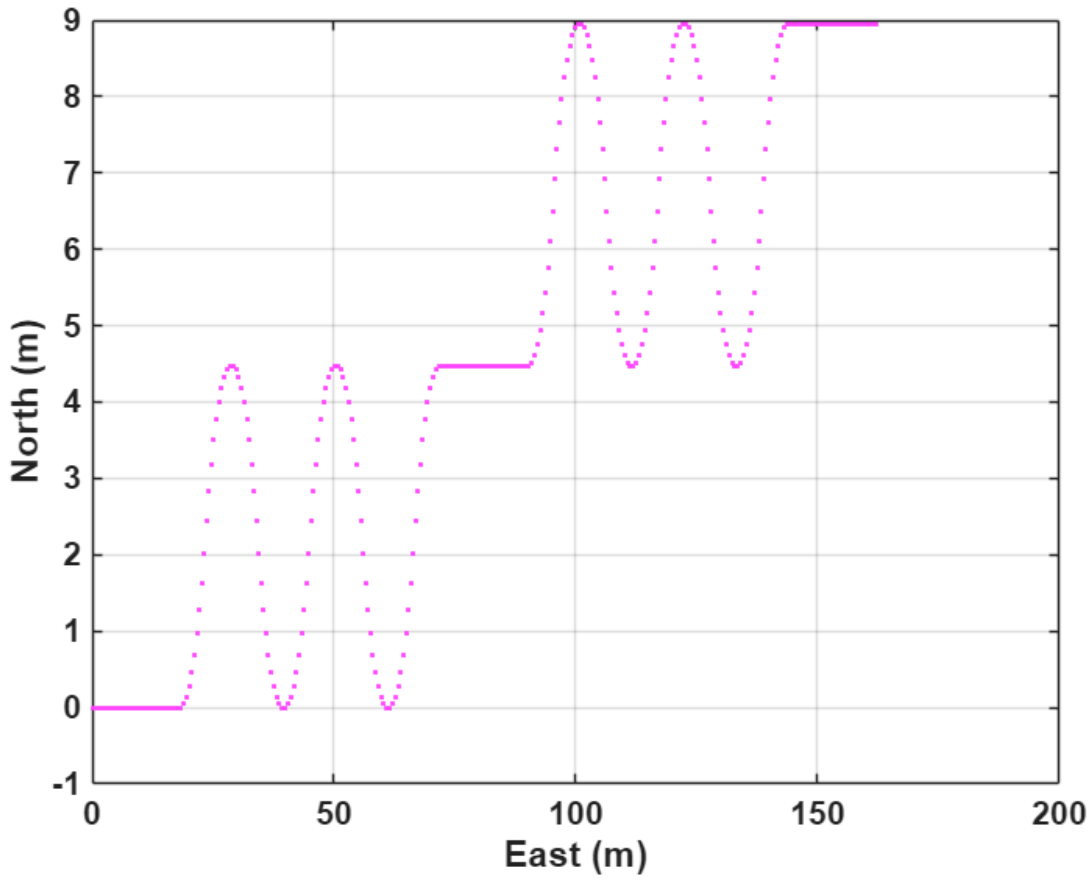


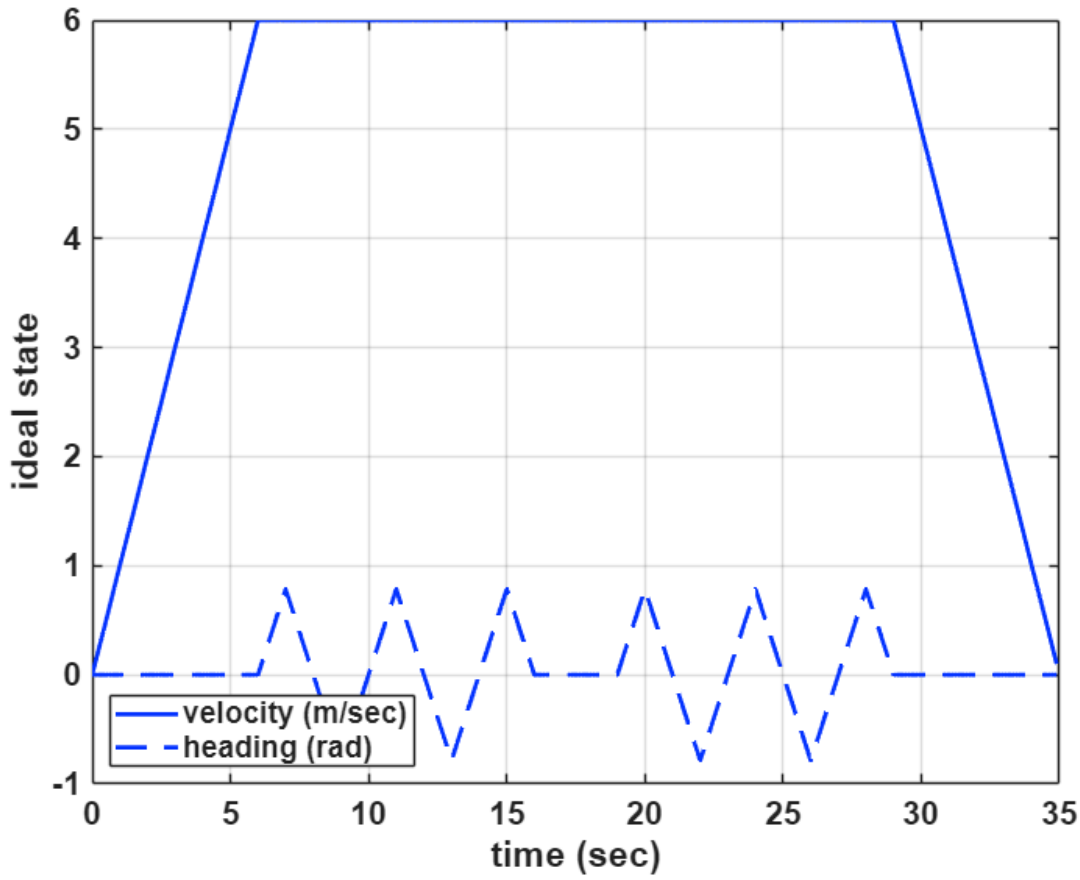

ideal state
time (sec)
velocity (m/sec)
heading (rad)

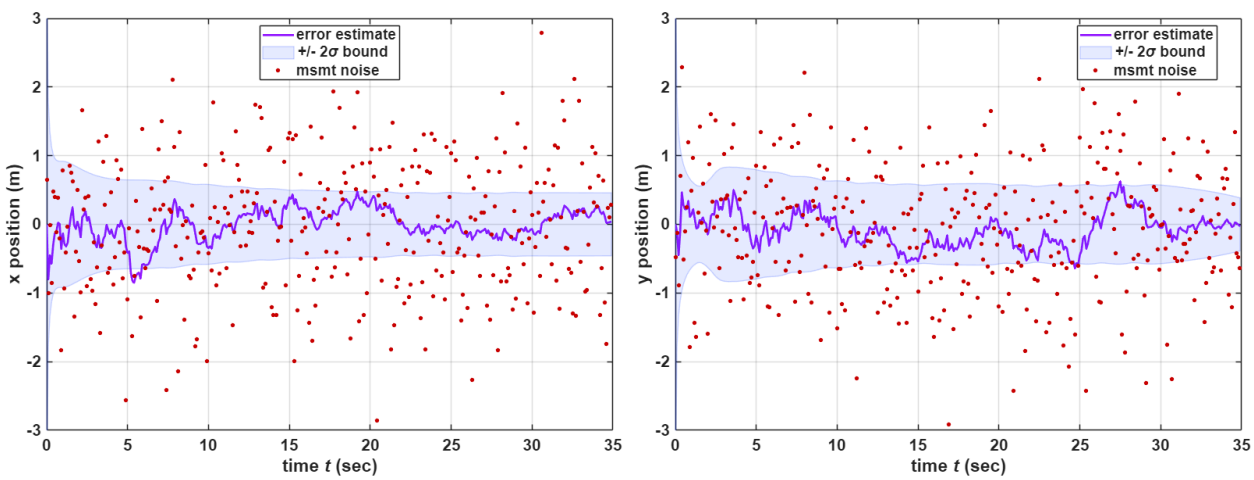

error estimate
+/- 2σ bound
msmt noise
x position (m)
y position (m)
time t (sec)

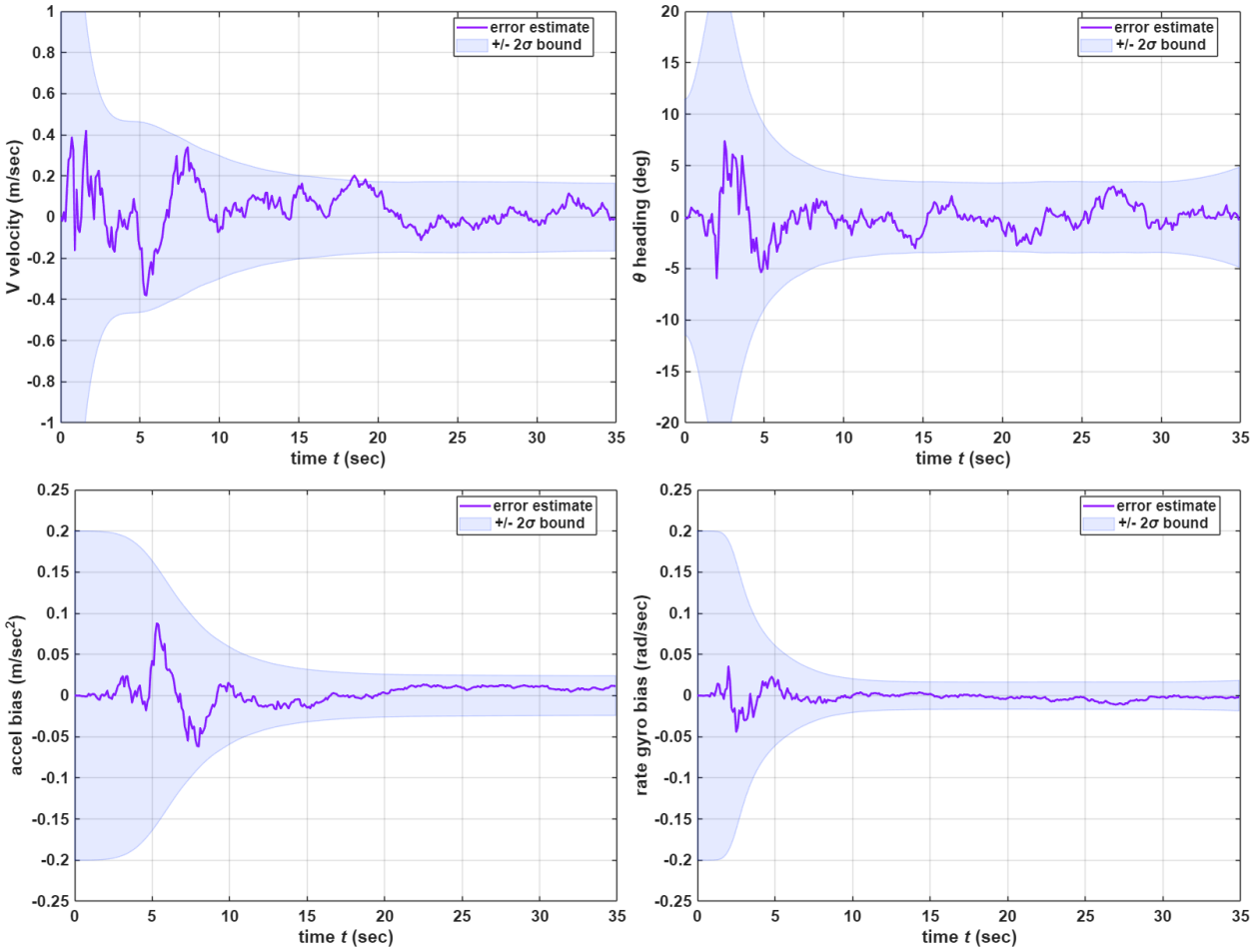

error estimate
+/- 2σ bound
V velocity (m/sec)
θ heading (deg)
accel bias (m/sec²)
rate gyro bias (rad/sec)
time t (sec)

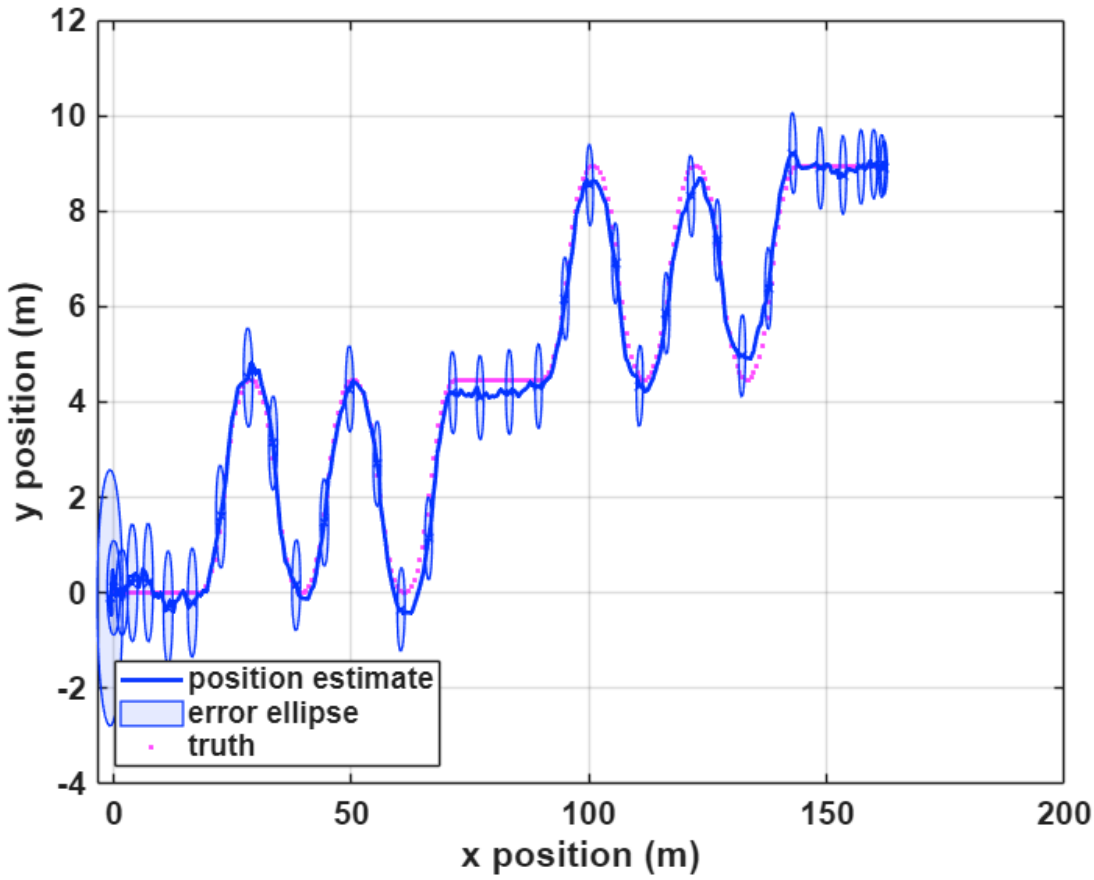

10
0
-10
-20
-30
-40
-50
y position (m)
-20
0
20
40
60
80
100
120
x position (m)
position estimate
error ellipse
truth

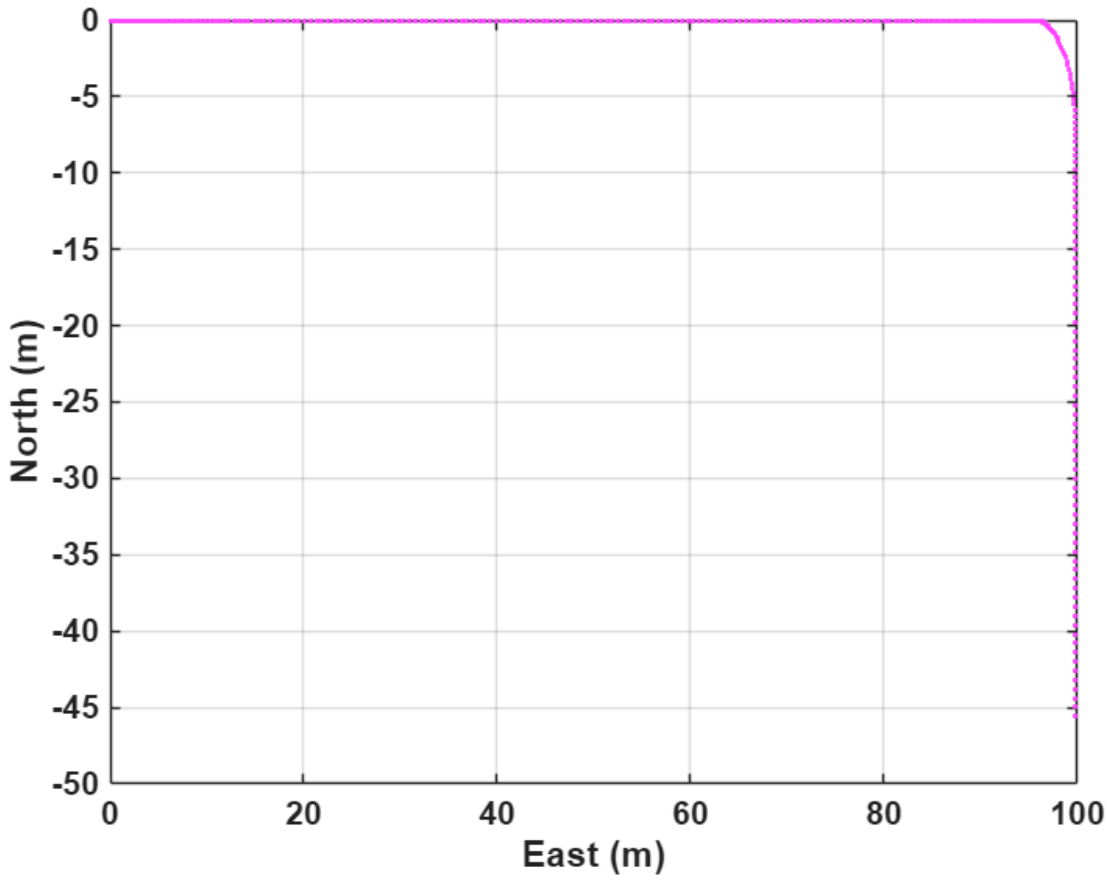

9
8
7
6
5
4
3
2
1
0
-1
North (m)
0
50
100
150
200
East (m)

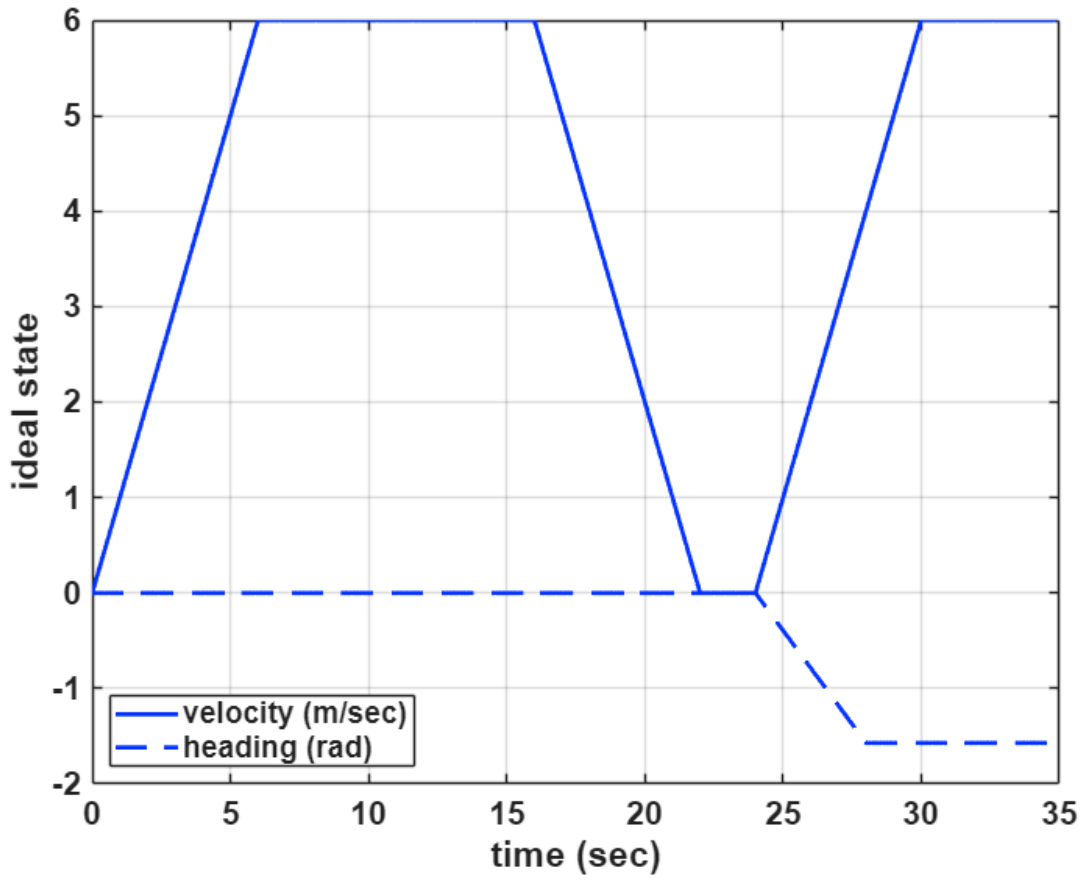

ideal state
time (sec)
velocity (m/sec)
heading (rad)

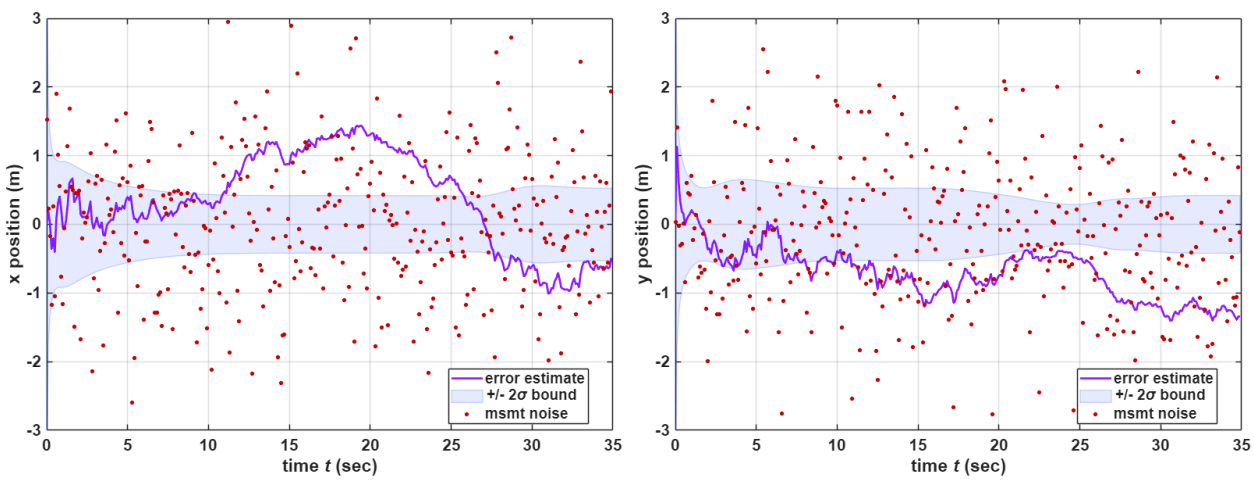

error estimate
+/- 2σ bound
msmt noise
x position (m)
time t (sec)
error estimate
+/- 2σ bound
msmt noise
y position (m)
time t (sec)

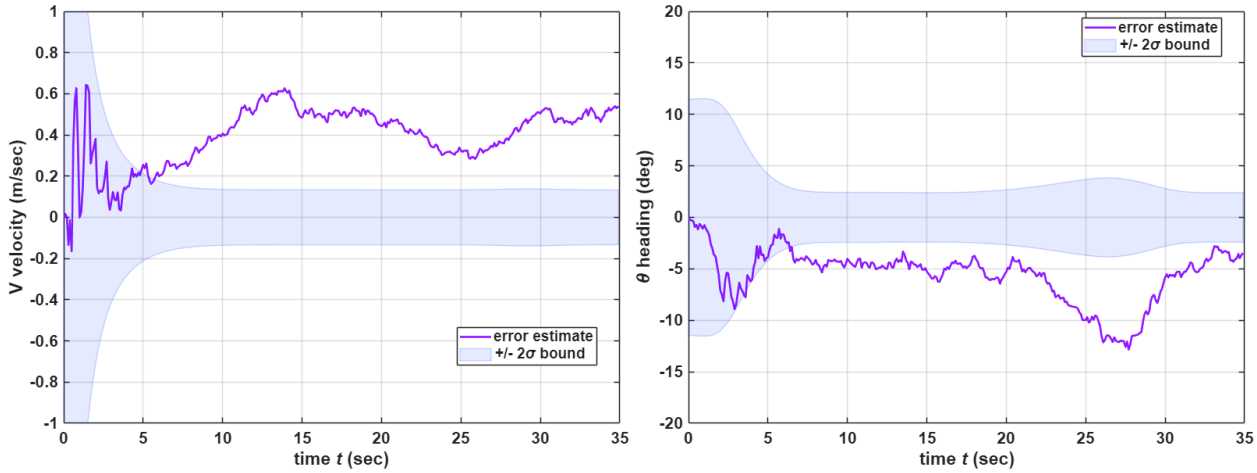

error estimate
+/- 2σ bound
V velocity (m/sec)
θ heading (deg)
accel bias (m/sec²)
rate gyro bias (rad/sec)
time t (sec)

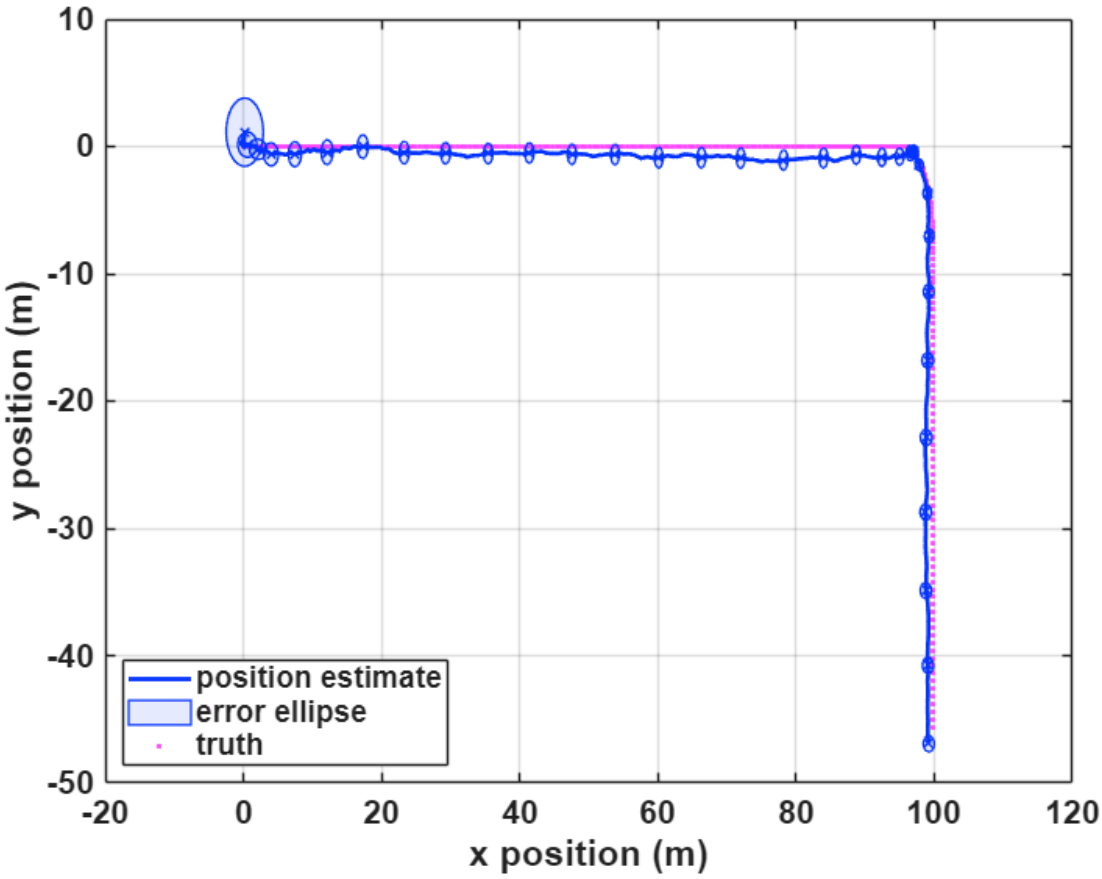

12
10
8
6
4
2
0
-2
-4
y position (m)
0
50
100
150
200
x position (m)
position estimate
error ellipse
truth

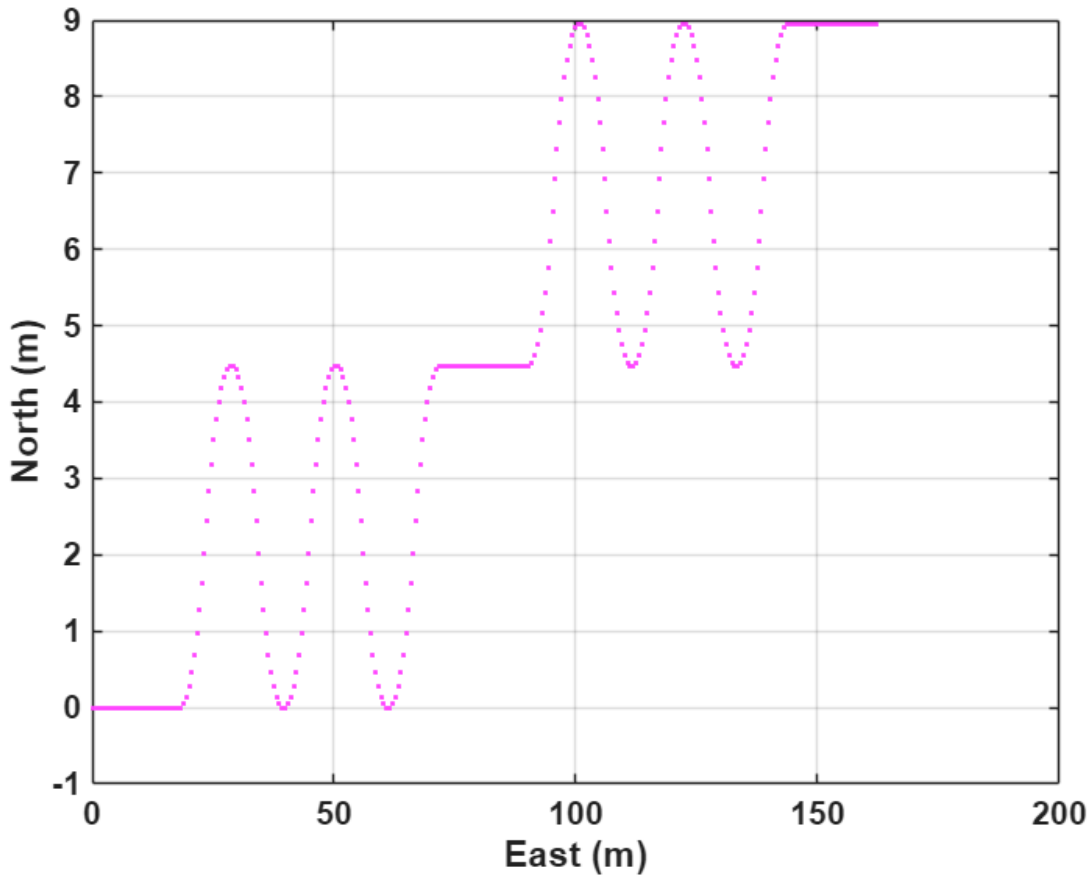

0
-5
-10
-15
-20
-25
-30
-35
-40
-45
-50
North (m)
0
20
40
60
80
100
East (m)

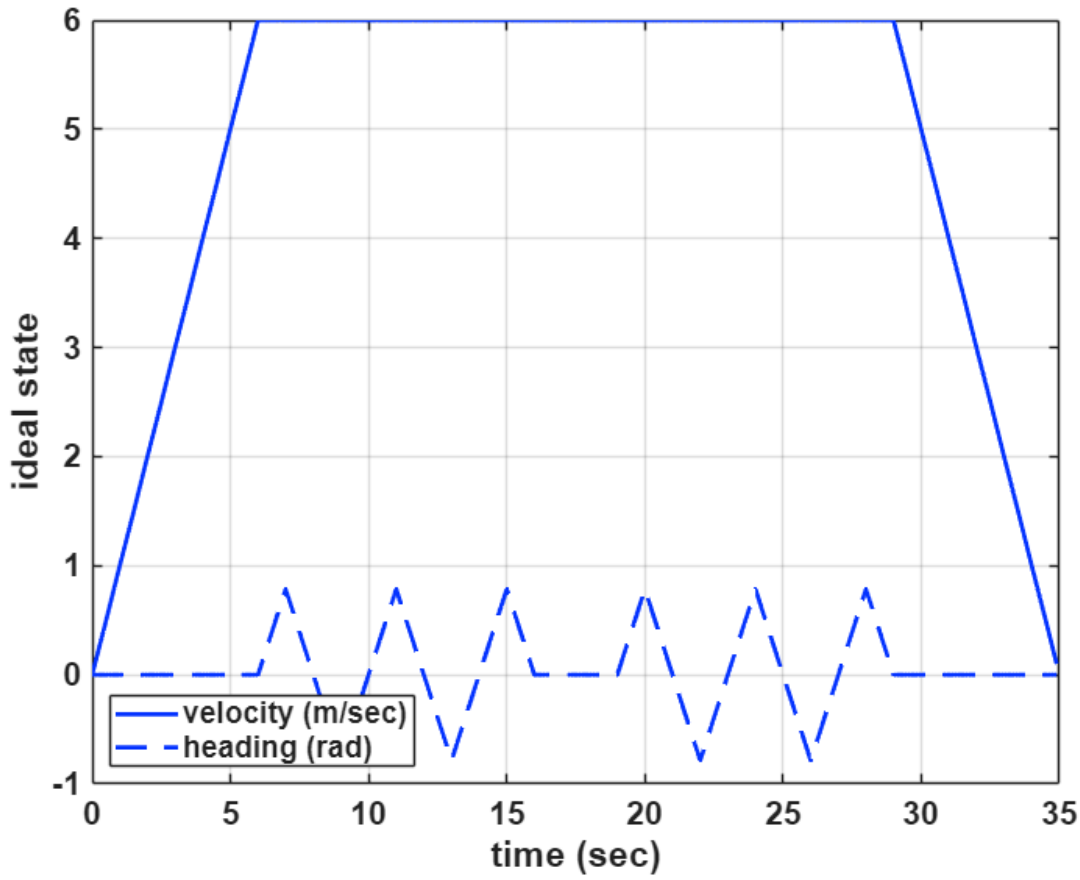

ideal state
time (sec)
velocity (m/sec)
heading (rad)

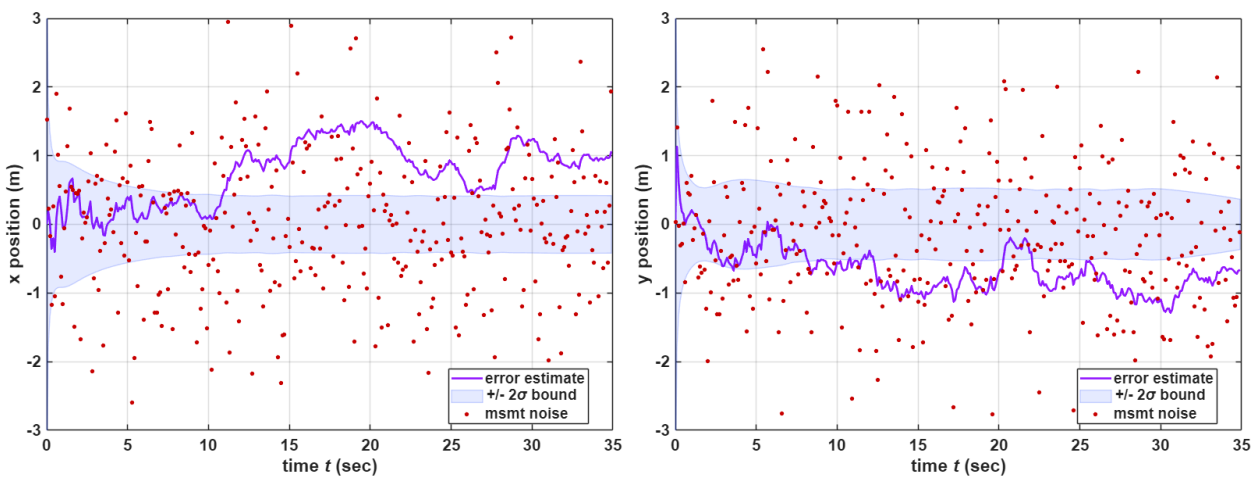

x position (m)
time t (sec)
error estimate
+/- 2σ bound
msmt noise
y position (m)
time t (sec)
error estimate
+/- 2σ bound
msmt noise

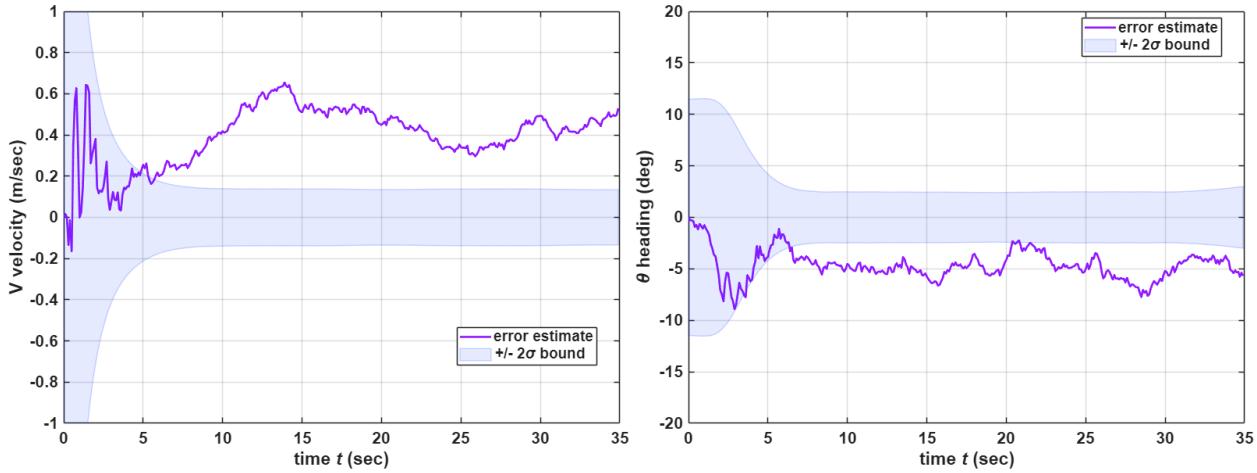

V velocity (m/sec)
time t (sec)
error estimate
+/- 2σ bound
θ heading (deg)
time t (sec)
error estimate
+/- 2σ bound

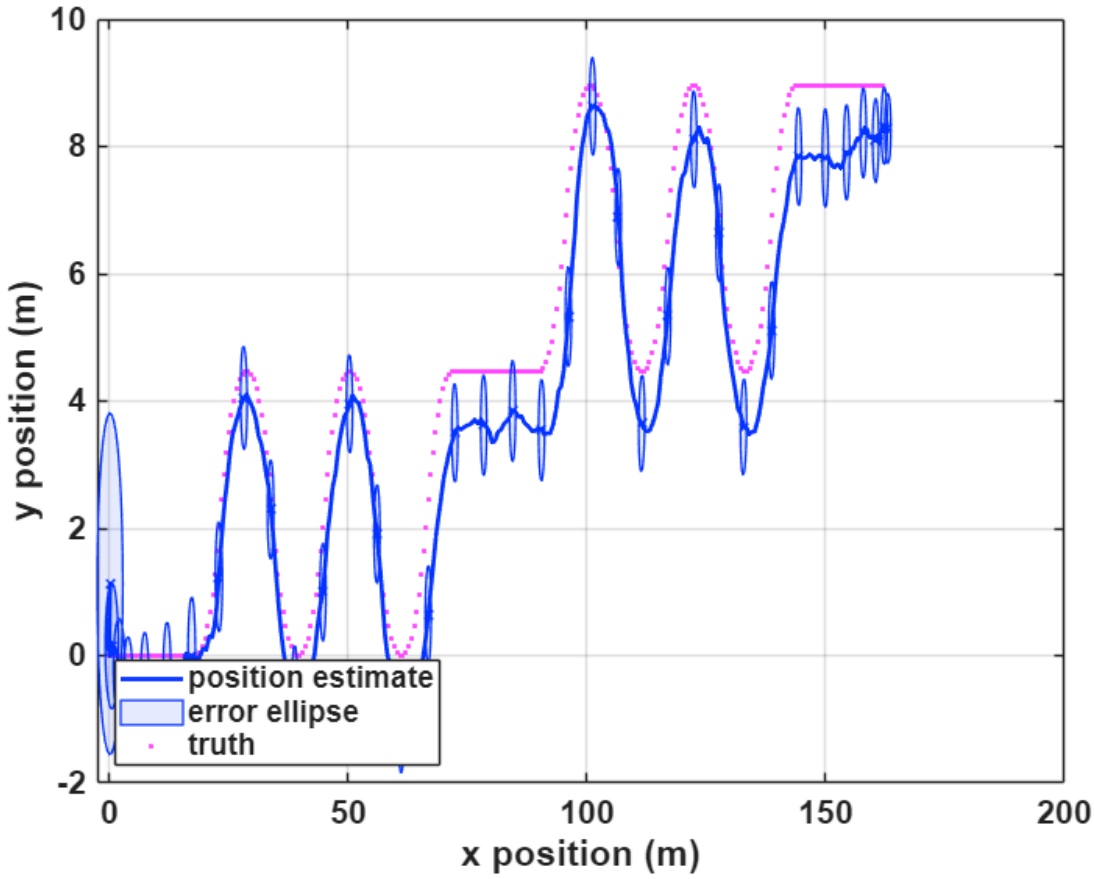

y position (m)
x position (m)
position estimate
error ellipse
truth


9
8
7
6
5
4
3
2
1
0
-1
North (m)
0
50
100
150
200
East (m)


ideal state
time (sec)
velocity (m/sec)
heading (rad)


x position (m)
y position (m)
time t (sec)
error estimate
+/- 2σ bound
msmt noise





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